

**FreshRoute - A Smart Fruit Supply Chain System to Reduce
Waste and Eliminate Middlemen in Sri Lanka**

25-26J-444

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology Specializing in
Software Engineering

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

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DECLARATION

I declare that this is my own work and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by any other person except where the acknowledgement is made in the text. Also, I hereby grant to Sri Lanka Institute of Information Technology, the nonexclusive right to reproduce and distribute my dissertation, in whole or in part in print, electronic, or other medium. I retain the right to use this content in whole or part in future works (such as articles or books).

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Date


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Date

ABSTRACT

Sri Lanka's value chain for fruits is highly inefficient with post-harvest losses of up to 20-40%. This institutional inefficiency is also catalyzed by the ongoing economic crisis in the country, which has trimmed farmer margins and raised the risks of food security. The root of this inefficiency lies in a pervasive information asymmetry where smallholder farmers take life-affecting harvesting decisions under extremely high uncertainty using outdated information and usurious go-betweens. Existing digital interventions in agriculture in the region are largely static informational websites, lacking predictive, actionable intelligence to support anticipatory decision-making. This proposal outlines "FreshRoute," an integrated, AI-driven decision-support system intended to serve as a bridge for this void. The basis of the system is a multivariate Long Short-Term Memory (LSTM) time-series model, built to boil down complex, non-linear interdependencies of historical market prices, meteorological data, and seasonal yield trends to generate accurate, real-time demand forecasts. Realizing the socio-technical challenges of deployment, these forecasts and actionable recommendations (e.g., "Delay Harvest") are made available through an offline-capable mobile app for low-connectivity settings supplemented with a multi-channel alarm system based on ubiquitous SMS and push notifications to bridge the digital divide. The primary contribution of this project is the development, implementation, and assessment of an end-to-end forecasting system that translates advanced data into practical value for smallholder farmers. The aim is to equip such farmers with information-based knowledge, reduce post-harvest losses, enhance resilience in supply chains, and provide superior economic yields to vulnerable agricultural communities.

Keywords: Demand Forecasting, LSTM, Smart Agriculture, AI in Supply Chain

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
AWS	Amazon Web Services
CIC	Cargills (Ceylon) PLC (A major agribusiness company in Sri Lanka)
DB	Database
FCM	Firestore Cloud Messaging
GCP	Google Cloud Platform
GPU	Graphics Processing Unit
HARTI	Hector Kobbekaduwa Agrarian Research and Training Institute
HTTPS	Hypertext Transfer Protocol Secure
IDE	Integrated Development Environment
JSON	JavaScript Object Notation
LSTM	Long Short-Term Memory
WAPE	Mean Absolute Percentage Error
ML	Machine Learning
MSE	Mean Squared Error
MVP	Minimum Viable Product
SARIMA	Seasonal Autoregressive Integrated Moving Average
SLIIT	Sri Lanka Institute of Information Technology
SMS	Short Message Service
SQL	Structured Query Language
SSL	Secure Sockets Layer
UAT	User Acceptance Testing
UI	User Interface
UX	User Experience
VM	Virtual Machine
WBS	Work Breakdown Structure

1 INTRODUCTION

The Sri Lankan agricultural scene is characterized by a relentless and destructive paradox: the great horticultural promise that is squandered through systemic ineffectiveness. A whopping 20-40% of all fruits that are picked are lost on the journey from farmer to consumer, a rate cited by the Hector Kobbekaduwa Agrarian Research and Training Institute (HARTI) that highlights a recurring breakdown of the national supply chain. Such loss is no figment of the imagination but is a reality that is concretely felt each day at the country's main centers of distribution such as the Dambulla Economic Centre where a conservative estimate of 8 tons of perishable produce is lost each day. Even though this quantity of physical loss has been a long-time problem in itself, the economic toll of that loss has been exacerbated to a very severe level through Sri Lanka's recent and grave economic crisis. The crisis has presented a perfect storm to the country's smallholder farmers.

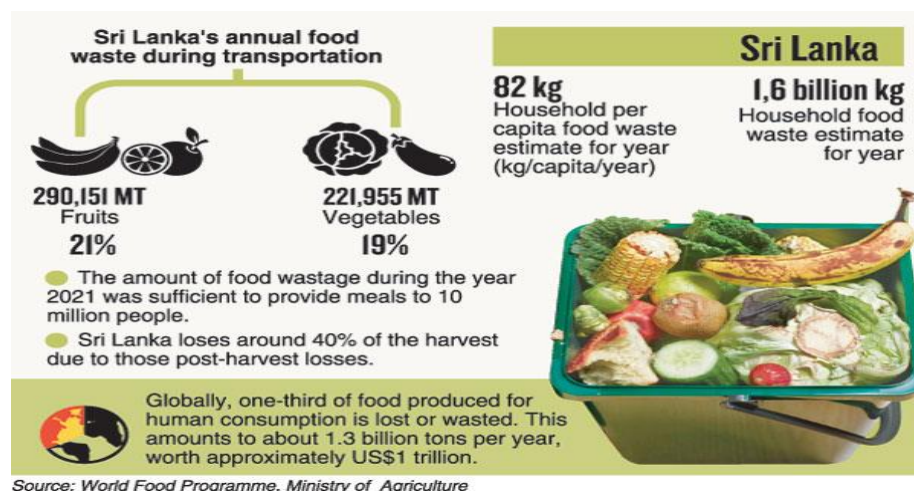


Figure 1: Sri Lanka's annual food waste

The country's economy experienced its highest-ever contraction in 2022, with the agricultural sector shrinking significantly. This macroeconomic collapse translated into hyperinflation of essential agricultural inputs. The price of a 50 kg bag of urea fertilizer, for instance, skyrocketed from LKR 3,050 to LKR 29,000, an increase of over 850%. Faced with unbearable production costs, farmers have been forced into defensive, often detrimental, adaptations. In the vital banana sector, cultivated acreage for some varieties dropped by 45%, with many farmers shifting to lower-yield, low-input varieties simply to remain solvent.

This combination of factors has engendered a vicious cycle. The production cost per unit of fruit is now markedly higher, and the overall potential output and revenue are lower. In this high-stress situation, each lost fruit after harvest is a crushing and exaggerated financial setback. The systemic inefficiency that long marred the farming communities has turned into immediate economic peril. Therefore, it is the contention of this study that reducing post-harvest loss is no longer an optimization exercise but a seminal intervention in developing economic resilience. By reducing waste through the intelligent matching of supply and demand, the system of FreshRoute seeks directly to offset the drastic margin compression caused by the economic crisis and offer a lifeline to smallholder farmers that is critically indispensable.

At the root of Sri Lanka's post-harvest loss predicament is a radical and universal information vacuum. Most fruit growers exist in a bleakly reactive and unenlightened decision environment with no access to reliable forward market information upon which to make the critical economic choice of when to harvest. They are therefore left to rely on no more than history, speculation, or the advice of intermediaries. These go-betweens have long profited from this informational edge and charged access to markets and prices accordingly. The financial crisis has exacerbated the imbalance further; a recent study of banana producers found that profit margins among producers fell by a median of 9.7% but increased by a median of 29.4% at the wholesale level. The result is a system in which value is being stripped out of but not created on the original producer.

Efforts to narrow this chasm using information and communications technology have so far been in vain. Current digital agric initiative in Sri Lanka like the government's "Govijana" mobile app and private enterprise "Govi Mithuru" portal are predominantly static information sites. These are "Level 1" digital instruments, suitable only to deliver historical information, e.g., market prices from last week, or disseminate generic, non-personalized agric advice. While valuable, they fundamentally do not respond to the farmer's most immediate and critical question: "What will the price and demand level be when my product is ready for harvest?" Such a passive approach to information delivery continues the very demand lag cycle that results in market surpluses and price spikes and ultimately wastage.

This concept introduces the FreshRoute system as a "Level 2" web-based intervention, a specifically designed one aimed at transcending the delivery of plain static information and providing predictive, actionable intelligence. The framework is supported by a core hypothesis: By creating a multivariate Long Short-Term Memory

(LSTM) model integrating meteorological, market, and seasonal information, and sharing its forecast results on a socio-technically appropriate platform, farmers can be empowered to align harvesting times with predicted demand, reducing waste and improving profitability. FreshRoute's innovation, however, is not only in the complexity of its algorithms but also in its user-centric delivery mechanism, one that is designed to break the paradox of digital intervention in emerging economies. Global research on applying AI in farming to the Global South consistently identifies key adoption impediments such as low digital literacy, sporadic or a lack of internet access in rural areas, and distrust of complex technology. A purely high-tech solution, such as a high-data smartphone application requiring constant connectivity, would consequently never reach a significant portion of its desired customer base. FreshRoute's solution addresses it head on. The reference to an "offline-capable mobile dashboard" and of supporting "SMS and push notifications" is not a laundry list of product features but a strategic, hybrid high-tech/low-tech solution. The SMS channel ensures critical, time-sensitive alarms are received by low-end feature phone-wielding farmers, the most ubiquitous technology in rural communities. The smartphone app's offline capability ensures that even low- or expensive mobile data users will get the latest synced forecast and recommendation. The dual-pronged approach is founded on the concept that in order to be revolutionary, technology must be accessible first. By virtue of a superior but highly integrated AI hub embedded into a functional and representative delivery framework, FreshRoute is seeking to breach the digital divide and deliver advanced, data-driven tools directly into the very hands of the most in-need farmers.

1.1 Background and Literature Survey

Effective demand forecasting is a cornerstone of modern agricultural supply chains, directly impacting profitability, resource allocation, and food security. Globally, AI-powered models have demonstrated significant potential in improving forecast accuracy, optimizing inventory, and reducing waste. Advanced techniques, particularly Long Short-Term Memory (LSTM) networks, have become a gold standard for time-series forecasting due to their superior ability to model complex, non-linear temporal dependencies [3],[6], [7] and capture long-term patterns that traditional statistical methods (e.g., ARIMA, SARIMA) often miss. Their proven efficacy in domains like stock market prediction, energy load forecasting, and retail demand planning provides a strong theoretical foundation for their application in agriculture.

Yet the application of these advances in technology to the particular agricultural and socio-economic environment of Sri Lanka is critically underexamined. While Ranaweera et al. (2023) observe that there is a notable lack of big-data ML-driven price forecasting research on Sri Lankan agricultural commodities [4], theirs is seminal in highlighting the climatic parameters in particular the rainfall and temperature locally as key determiners of price variability, highlighting the extreme effect of unpredictable meteorological factors on market instability [10]. We are consequently left with the task of not only forecasting but of modeling a highly complex multi-variable system where market forces and climatic conditions meet.



Figure 2: key factors influencing digital agricultural production

At the same time, the field of development informatics has proven that technology's effect is not only determined by the power of algorithms but also by its accessibility. Previous studies, including systems implemented in nations such as Ethiopia and India, verify the success of mobile and SMS-enabled systems in providing information of substance to rural communities with high usability and coverage [11]. These confirm that low-bandwidth and high-penetration technologies can frequently be more successful in overcoming the digital divide compared to advanced applications.

In spite of these other threads of work in the literature, a critical gap remains in literature. Current work in Sri Lanka is typically compartmentalized: either it is academic demonstrations-of-concept for forecasting with a weak delivery mechanism [2], [9], or they are generic information sites providing static, historical information without predictive, actionable intelligence (e.g., Govijana app). There is no integrated solution that synergistically brings together an advanced, multivariate LSTM-powered

forecasting engine specially calibrated for the Sri Lankan fruit crops and their distinctive climatic drivers with a practical multi-channel (SMS + push notification) alerting system tailored to the realities of smallholder farmers. There is a corresponding effort that aims to bridge this critical gap with a whole-systems approach in this project named FreshRoute that brings advanced data science to tangible, accessible decision-support for the most in-need farmers.

1.2 Research Gap

1.2.1 Related work

There are various foreign and domestic research that discussed the utilization of information systems and machine learning in agriculture, but a significant knowledge gap remains in developing an integrated AI-driven demand forecasting and decision-support system for the Sri Lankan fruit supply chain. De Silva created a whole mobile-based information system to enhance farm operations in Sri Lanka and recognized the role of mobile technologies in the provision of agricultural information, but the study lacked strong predictive analytics and in-real-time forecasting [1].

In the same line of thought, Ranaweera et al. applied crop price forecasting of the Sri Lankan vegetable market with the aid of machine learning methods and showed regional relevance but only focused on vegetables and didn't consider the feature of real-time warning mechanisms [4]. Wanigasinghe and Ranasinghe proposed a neural network system for Sri Lankan economic centers, but it didn't involve state-of-the-art deep learning methods like LSTM nor environmental factors like weather and seasons [9].

Aphinaya et al. utilized the SARIMA model in the prediction of the prices of jackfruit and validated the relevance of time-series techniques though conventional statistics has difficulty with nonlinear and multivariate datasets [2]. At the global level, Yoo and Oh utilized Seasonal Long Short-Term Memory (LSTM) in agricultural sales volume forecasting and verified the efficacy of the model in capturing seasonal time-series information though the study lacked being applied to the Sri Lankan scenario and incorporated into farmer-facing systems. All the above studies confirm the promise of mobile and AI technologies in

support of agriculture though present fundamental shortcomings include the emphasis on backward-looking analysis, lack of forecasting in a live system, lack of integrated warning systems, and narrow coverage of fruit supply chains [3]. The present study compensates for the shortcomings through the development of a live forecasting module of the LSTM kind with SMS and push messages specially designed for Sri Lankan fruit farmers.

1.2.2 Problems with using the existing system

Though there are a number of digitization programs in Sri Lanka, incumbent systems have considerable shortcomings that preclude them from making a meaningful contribution to solving actual concerns of fruit producers. The majority of systems are static and descriptive systems, e.g., the Department of Agriculture's Govijana app and Dialog's Govi Mithuru, offering mainly historical market prices and typical weather information without forecasting ability. Such a reactive form inhibits the ability of farmers to make strategic decisions about future demand and prices, and the repetition of the cycle of oversupply and price collapse continues. Other systems lack tailored, actionable intelligence and present generic advisories instead of clear recommendations for particular crops, geographies, or harvesting seasons.

Further, the coverage and connectivity of incumbent systems are limited to some of them offering isolated information without connectivity to marketplaces or logistics support chains, and the other ones, e.g., CIC's in-house tools, are integrated into proprietary ecosystems that are inaccessible for the majority of smallholders. Research activity from academia is promising but stays at the level of proof-of-concept studies without transferring to the realities of the field

Secondly, technology and accessibility restrictions rule out the majority of farmers with simple phones and sporadic internet connectivity because the majority of the applications presume perpetual connectivity and do not offer offline or SMS-based fallback options.

Thirdly, adoption of AI-enabled forecasting is notable in the absence of the same in the solutions because although simple rule-based alarms exist, none utilize superior time-series models like LSTM that could take into account past prices, weather, and seasonal patterns and produce reliable proactive demand forecasts. All of the gaps put together create a compelling necessity for a smarter, accessible system that is farmer friendly.

Table 1: Problems with using the existing system

Category	Limitation	Examples	Impact on farmers
Static and Descriptive Nature	Focus only on past and present data; no predictive forecasting	Govijana app, Govi Mithuru	Farmers make decisions based on outdated information, leading to oversupply and price crashes
Absence of Personalized, Actionable Insights	Broad, generic messages without crop/location-specific recommendations	CIC advisory tools	Farmers lack tailored guidance (e.g., when to harvest, how to respond to local price trends)
Limited Scope and Integration	Not linked with marketplaces, logistics, or broader supply chains	Govi Mithuru, CIC closed ecosystem, academic SARIMA models	Smallholders cannot access direct markets or scalable tools, restricting adoption
Technological and Accessibility Barriers	Require smartphones and constant internet; no robust offline or SMS functionality	Most mobile apps in agriculture	Excludes rural farmers with basic phones and poor connectivity
Lack of AI-Driven Forecasting	No advanced time-series or multivariate forecasting models (e.g., LSTM)	All current local systems	Farmers lack proactive intelligence to anticipate demand and prevent losses

1.2.3 Research Questions

➤ **Primary Research Question:**

- How can an integrated, AI-driven decision-support system reduce post-harvest waste and improve economic outcomes for fruit farmers in Sri Lanka by mitigating information asymmetry?

➤ **Secondary Research Questions:**

- To what extent can a multivariate LSTM model, trained on historical market prices, meteorological data, and seasonal yield patterns, accurately forecast short-term fruit demand in the Sri Lankan market?

- How is the predictive of an LSTM model compared to traditional time-series models (e.g., SARIMA, Prophet) for this specific agricultural context?
- What is an effective system architecture for collecting, processing, and serving real-time demand forecasts in a resource-constrained environment like rural Sri Lanka?
- How can a multi-channel notification system (SMS and push notifications) be optimally designed to ensure timely and reliable delivery of critical alerts to farmers with varying levels of technological access?
- What is the critical user interface (UI) and user experience (UX) design principles for an offline-capable mobile application that is accessible and actionable for farmers with potentially low digital literacy?
- What are the perceived barriers and facilitators to the adoption of such an AI-based forecasting tool among the target population of smallholder fruit farmers?
- How do farmers' harvest and sales decisions change when provided with data-driven demand forecasts and actionable recommendations?
- What is the potential economic impact of using the FreshRoute system, as measured by estimated reductions in post-harvest losses and improvements in profit margins for pilot users?

2 RESEARCH PROBLEM

The Sri Lankan fruit producers face a situation of extreme information inadequacy with limited capacity for predictive market demand volatility through technology. Such extreme lack of agricultural decision-support mechanisms translates into essentially wrong-timed harvesting strategies with colossally economic implications at the farm and national levels. Grand research questions seek to explore developing, designing, and implementing a useful system of real-time prediction and warning allowing farmers to strategically schedule their harvesting agendas in line with predicted market demand patterns. Such a complex problem demands the resolution of a range of interconnected questions:

Crop	Operations					Total loss in farm operations	Storage Channels					Total loss in storage	Overall Total Loss
	Harvesting	Collection	Sorting/Grading	Packaging	Transport		Farm	Godown/cold store	Whole-saler	Retailer	Processing unit		
Apple	4.33 ±0.35	0.29 ±0.22	3.94 ±0.44	0.11 ±0.11	0.42 ±0.19	9.08[#] ±0.30	0.02 ±0.00	0.13 ±0.02	0.57 ±0.08	0.34 ±0.29	0.25 ±0.11	1.31 ±0.11	10.39[*] ±0.24 (12.26)
Banana	1.62 ±0.35	0.26 ±0.14	2.06 [*] ±0.37	0.19 ±0.31	1.91 ±0.40	6.04 ±0.33	0.03 ±0.01	0.08 [#] ±0.04	1.16 ±0.35	0.45 ±0.07	0.00 [*] ±0.00	1.72 ±0.17	7.76 ±0.29 (6.60)
Citrus	1.68 ±0.48	0.33 ±0.10	3.71 ±0.59	0.18 ±0.13	1.65 ±0.24	7.55[*] ±0.38	0.04 ±0.01	0.02 [*] ±0.01	0.91 ±0.13	1.12 ±0.20	0.06 ±0.07	2.14[~] ±0.11	9.69[~] ±0.29 (6.38)
Grapes	1.77 ±0.2	0.30 ±0.05	3.36 ±0.42	0.10 ±0.06	0.98 ±0.18	6.52 ±0.26	0.01 [#] ±0.00		0.78 ±0.14	1.24 ±0.17	0.09 ±0.03	2.11 ±0.15	8.63 ±0.22 (8.30)
Guava	5.33 ±1.78	0.31 ±0.28	4.95 ±1.86	0.09 ±0.20	1.21 [#] ±0.64	11.90 ±1.24	0.23 ±0.06		1.62 ±0.52	2.08 ±0.73	0.04 [#] ±0.03	3.98± 0.55	15.88 ±1.11 (18.04)
Mango	2.09 ±0.71	0.30 ±0.44	3.26 ±0.77	0.23 ±0.07	1.04 ±0.14	6.92[#] ±0.54	0.11 ±0.01	0.01 ±0.01	0.69 ±0.39	1.18 ±0.42	0.25 ±0.09	2.24 ±0.28	9.16[*] ±0.50 (12.74)
Papaya	0.98 ±0.45	0.42 ±0.15	1.46 ±0.47	0.34 ±0.06	0.92 ±0.39	4.12 ±0.37	0.05 ±0.02	0.01 [*] ±0.00	0.79 ±0.17	1.71 ±0.25	0.03 [*] ±0.01	2.58± 0.12	6.70± 0.26 (7.34)
Sapota	2.53 ±1.18	0.35 ±0.07	2.55 ±0.56	0.28 ±0.17	1.70 ±0.48	7.41[*] ±0.67	0.01 ±0.01	0.25 [*] ±0.05	0.89 ±0.20	1.13 ±0.13	0.03 [*] ±0.02	2.31[*] ±0.09	9.73[*] ±0.46 (5.77)

^{*} Estimated losses are significantly higher in comparison to losses observed in 2005-07(previous study).

[#] Estimated losses are significantly lower in comparison to losses observed in 2005-07(previous study).

Figures in parenthesis represent the losses of previous study 2005-07.

Figure 3: Harvest and post-harvest losses (%) of fruits at the national level

Secondly, the forecasting and data integration task is to find the optimal method of integrating heterogeneous data streams like historical market prices, future weather forecasts, seasonal yield patterns, and perhaps social and economic information into accurate, trustworthy demand forecasts. It comprises developing and ideally calibrating separately the time-series analysis models individually at the micro level, that is, LSTM neural networks, in a way that one becomes able to capture the fine-grained time- and nonlinear agricultural market and Sri Lankan-particular climatic and economic phenomena interactions.

Second is the challenge of decision support delivery in transforming raw predictive output into actionable intelligence to be presented to rural agricultural communities. It is the creation of simple and consumable visualization interfaces through mobile dashboards and design robust multi-channel alert systems (through SMS and push) that will be useful in low connectivity scenarios. The system must be able to deliver not just predictions but also contextually relevant, farmer-level recommendations relevant to the specific situations of the communities, crops, and harvesting status.

Third is the access problem of how to deliver these new technology solutions in an accessible and useful format to low-digital literacy farmers with sporadic internet access and uneven access to technology. The system needs to be capable of being able to support adequate offline capability, low data requirements, and easy interface that will be able to advance the digital divide without compromising the ability of prediction and timely responses.

Addressing the above issues will facilitate a paradigm shift from reactive guessing to proactive intelligence-driven farm planning. By correlating the harvesting decisions with the actual market-purchasing pattern of consumers rather than . The study hopes to minimize post-harvest loss significantly, maximize supply-demand matching the market level, and maximize farmer profitability across Sri Lanka's agricultural supply chain significantly.

3 OBJECTIVES

3.1 Main Objective

- ❖ It is necessary to design, develop and deploy an integrated real-time demand forecasting and alert system fueled by artificial intelligence that helps Sri Lankan fruit farmers make the most of Harvest planning and sales decisions through informed market demand forecasts that are data-driven. The system will employ the latest time-series machine learning techniques, particularly the Long Short-Term Memory (LSTM) networks, to analyze disparate sources of data and generate actionable information that results in reducing post-harvest losses, maximizing profits and supply chain efficiency optimization in Sri Lanka's fruit agriculture industry segment.

3.2 Specific Objective

The project will gather historical fruit market price and demand data, alongside weather and seasonal yield records, from trusted sources. This data will then undergo cleaning to remove errors, normalization for consistency, and feature engineering to add meaningful patterns, ensuring a reliable dataset suitable for accurate model training and forecasting.

- ❖ A time-series model, LSTM, will be designed, trained, and evaluated using the prepared datasets to forecast short-term fruit demand. Their predictive performance will be compared with statistical metrics such as Mean Squared Error. This process ensures selection of the most effective forecasting approach for practical deployment in the system.
- ❖ An offline-capable Android mobile dashboard will be developed to display forecasted demand, visualize the impact of weather, and offer clear harvest timing recommendations. The dashboard will support local languages, operate smoothly in rural environments with poor connectivity, and present insights in an accessible, farmer-friendly format for effective decision-making.
- ❖ A real-time alert system will be implemented to provide timely notifications of demand shifts and harvest advisories. Farmers with basic phones will receive SMS alerts, while smartphone users will get push notifications. This dual-

channel design ensures inclusiveness and guarantees that users can respond quickly to changing market conditions.

- ❖ The forecasting module will be integrated into a larger supply chain platform, enabling future scalability and broader adoption. Predictions will be validated against actual market outcomes to ensure trust and accuracy. This integration strengthens market linkages, provides transparency, and enhances farmers' decision-making power through reliable, data-driven insights.

4 METHODOLOGIES

4.1 High level Architecture

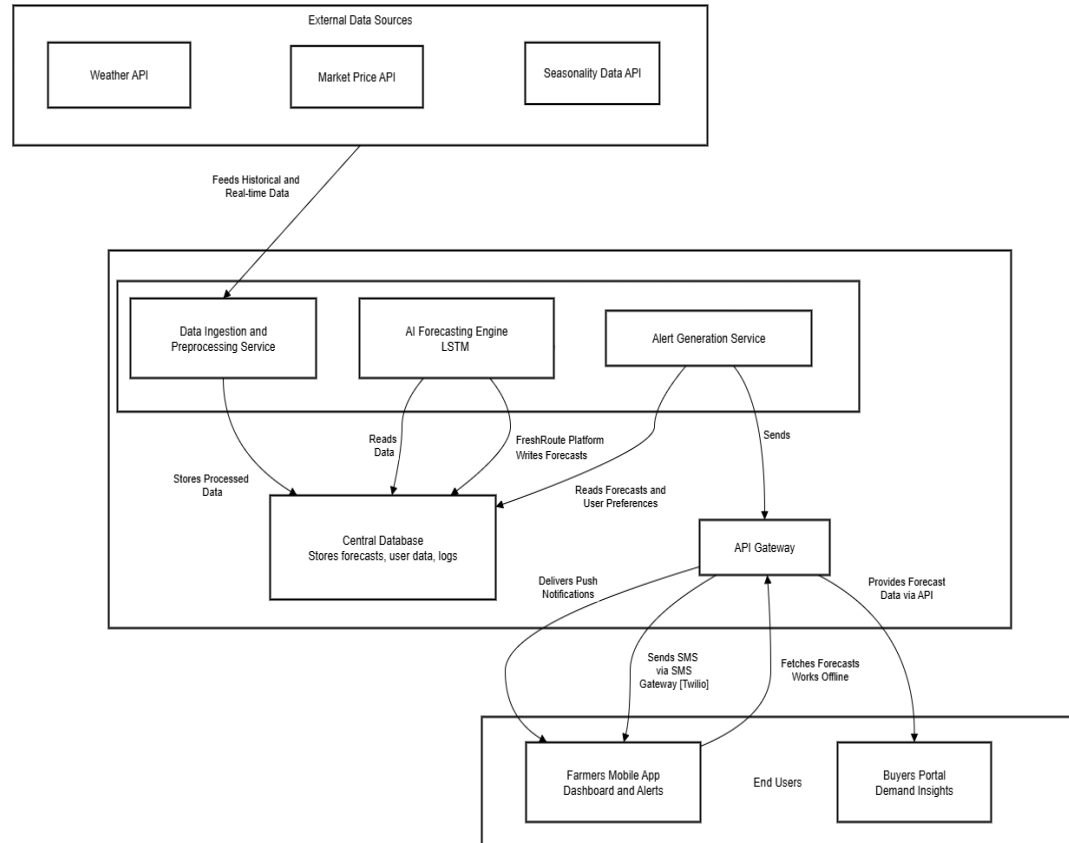


Figure 4: High level Architecture

This architecture diagram outlines the end-to-end flow of the FreshRoute Demand Forecasting System. External sources of data from weather, market, and seasonal APIs are ingested and preprocessed through a specialized service. The main AI Forecasting Engine using an LSTM model processes this information to produce reliable demand forecasts. These forecasts are saved in a central database. An Alert Generation Service continuously surveys forecasts and client preferences and initiates live SMS and pushes notifications through Twilio and Firebase in real-time. At the same time, an API Gateway delivers forecast information to a mobile application for farmers with a design for offline access and a buyer-side web portal that provides easy access to vital information.

4.2 System Overview Diagram

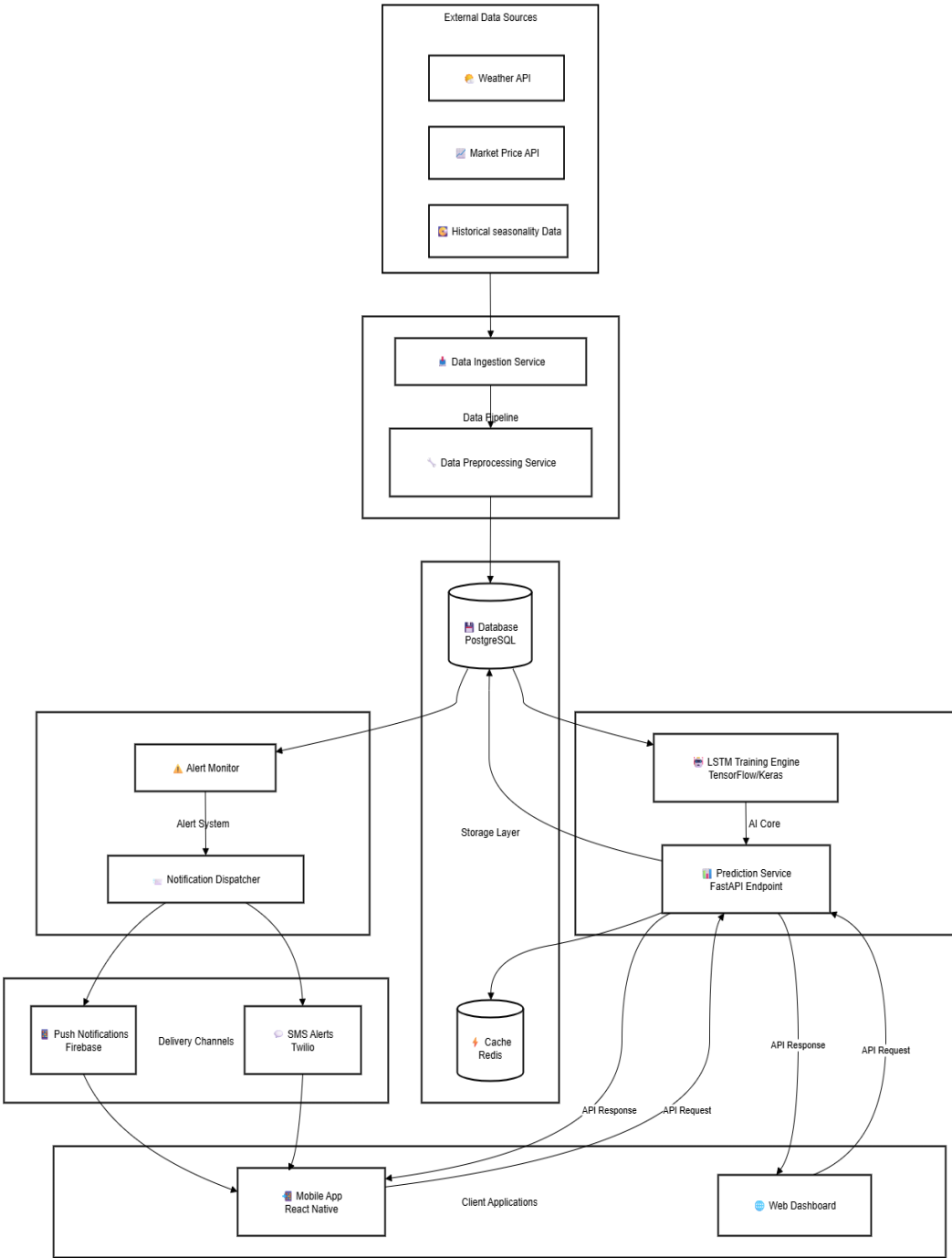


Figure 5: System Overview Diagram

This design represents the end-to-end flow of the demand forecasting system. External data is ingested, preprocessed, and stored in PostgreSQL. The AI core utilizes an LSTM model for training and a FastAPI service for forecasting. Output is caught in Redis for performance. The system is always keeping track of forecasts; when it discovers dramatic market changes, the alert dispatcher sends multi-channel notices through Firebase and Twilio. At the same time, both the mobile and the web client consume detailed forecasting data through API requests so that the farmers get the real-time, actionable information directly into their devices so that they can make data-driven harvest and sales decisions.

4.3 Data Collection

- ❖ **Data Sources:** The system under consideration is reliant on diverse and abundant sources of data in order to achieve reliable and accurate forecasting. Historical market prices will be scraped from government agricultural department websites, economic center portals such as the Dambulla Economic Centers of Sri Lanka, and market board reports, providing a general overview of the trend of prices. Weather data will be acquired through APIs from OpenWeatherMap and the Department of Meteorology Sri Lanka and contain historical records and forecasted scenarios such as temperature, rains, and humidity, effectively regulating the demand and supply of crops significantly. In addition to that, seasonal yield data will be gathered from the annual reports released by the Hector Kobbekaduwa Agrarian Research and Training Institute (HARTI) and the Department of Census and Statistics and provide valuable information regarding production trends and seasonal shifts. These datasets collectively form the foundation of the training and deployment of the LSTM-based forecasting model.
- ❖ **Data Preprocessing:** The preprocessing phase plays a foundational role in the development of high-quality model-ready datasets from the raw inputs. The preprocessing starts with cleaning where the missing values are handled through interpolation methods or in the event of faulty records discarded altogether in order to ensure data integrity. Normalizing is then done to transform numerical variables like prices and temperatures into a consistent range (i.e., 0 through 1), improving the stability and performance of the LSTM model. Feature engineering is then done to enrich the dataset through the addition of other meaningful attributes like rolling averages of prices in order to capture short-term trends, cyclical demand patterns through the addition of seasonal indicators like month or festival period, and temporal dependencies through the addition of lag feature like the price of the

preceding week. All in all, the preprocessing steps stand to keep the data clean and contextually enhanced in order to achieve credible forecasting.

4.4 Component Methodology

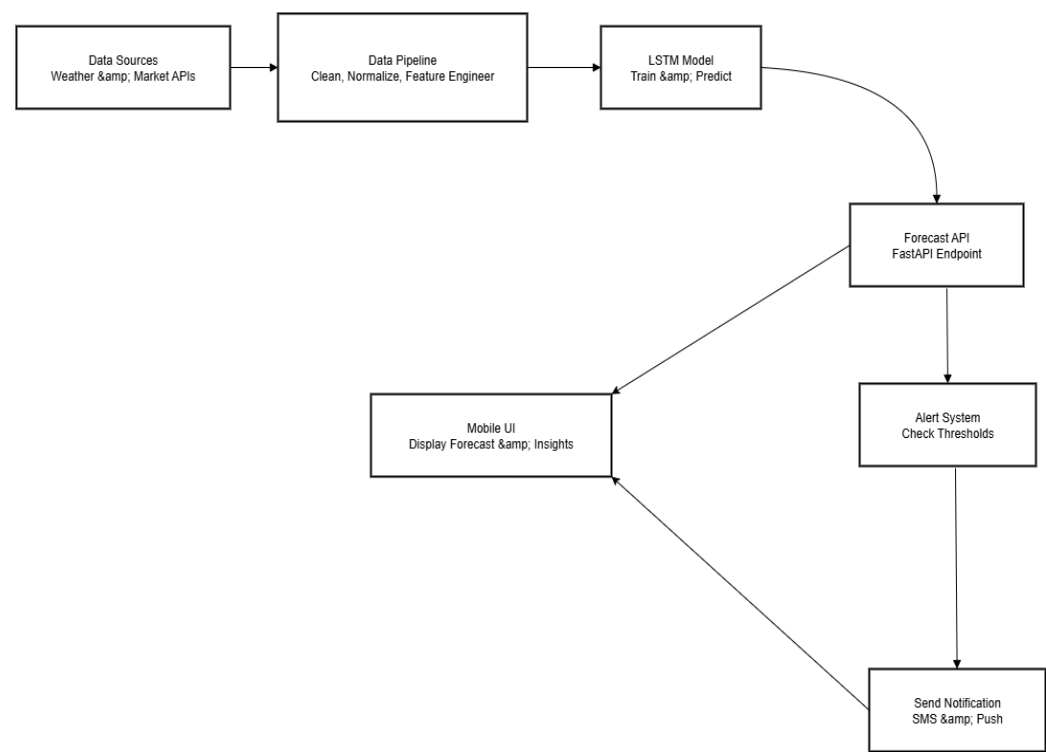


Figure 6: Component Methodology Diagram

The development and operation of the demand forecasting module will follow a structured, multi-stage methodology for data and model management. Each component has a specific role in ensuring the accuracy and reliability of predictions.

4.4.1 Data Acquisition & Preprocessing:

The foundational phase is dedicated to transforming raw data into clean, structured form ready for the LSTM model to be trained on. The data starts with data ingestion where Python scripts automatically pull information from external REST APIs (e.g., market prices and weathers), internal databases of historical yields, and so on. Then data cleaning is done where missing values are handled using interpolation methods and the statistics-based approach of Z-scores to get rid of the noise and achieve high-quality data. After that, the data is subjected to transformation where numerical features (e.g., price and temperature) are normalized into the $[0, 1]$ range using Scikit-learn's MinMaxScaler in order to aid in stable model learning. Feature engineering is also done where useful predictor variables are created like rolling averages (7-day average prices), lag variables (same week prices of the previous year), and seasonal indicators (month and holiday flags), which aid the model in uncovering hidden patterns and seasonality. The data is ready for supervised learning through the creation of a sequence where the tabular data is converted into time-series samples. Given a sequence of a size of 60, the inputs (X) are the prior 60-day observations, and the output (y) is the demand/price of the 61st day and this allows the LSTM to extract the temporal dependencies easily.

4.4.2 Model Development & Training:

This component is dedicated to building and refining the forecasting engine by designing an effective LSTM-based model. A Sequential model will be constructed using TensorFlow/Keras, beginning with an input layer shaped to accommodate the defined sequence length and number of features. The core of the architecture will include one or two LSTM layers with 50–100 units, enabling the model to capture complex temporal dependencies in the data. To reduce overfitting, dropout layers will be added after the LSTM layers, randomly disabling a fraction of units during training. The network will conclude with a dense output layer containing a single neuron with a linear activation function, designed to predict continuous values such as demand or price. For training, the model will employ the Adam optimizer alongside Mean Squared Error (MSE) as the loss function, ensuring robust and efficient optimization. A validation set will be used to monitor model performance, while an EarlyStopping callback will be applied to halt training when validation performance ceases to improve, thereby achieving an optimal balance between accuracy and efficiency [6],[7].

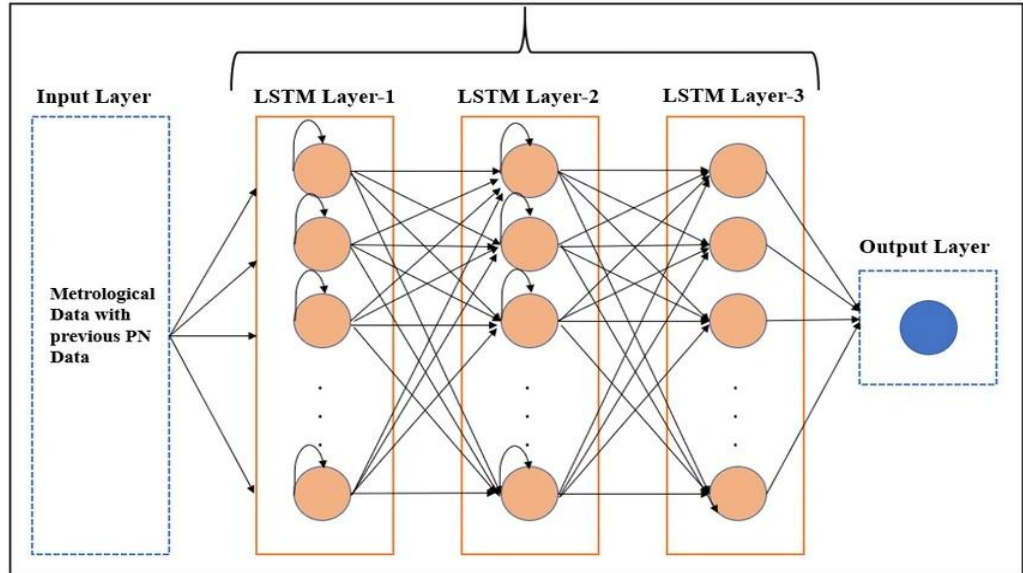


Figure 7: Basic LSTM Model Development Structure

4.4.3 Operational Prediction & Serving:

This describes how the trained LSTM model is deployed and utilized within the live system to generate real-time forecasts. The trained model, saved in a .keras file, along with the fitted scaler stored as a .pkl file, will be integrated into a **FastAPI-based service**, exposing a REST API endpoint (e.g., POST /predict) for external applications to request predictions. When a prediction request is received, the **inference workflow** begins by fetching the most recent historical sequence required for input. This sequence is then preprocessed using the saved scaler to ensure consistency with the training data. The preprocessed input is passed to the trained LSTM model, which generates the forecast. The raw prediction is subsequently inverse transformed back to the original scale for interpretability, and the system returns the final forecasted value along with an associated confidence metric, ensuring reliable and actionable predictions in real-world operations.

4.4.4 Integration & Alerting:

This component focuses on delivering critical insights to users in a timely and actionable manner. A dedicated **Alert Service** continuously monitors new forecasts stored in the database and evaluates them against user-defined thresholds to identify significant market shifts or anomalies. When such conditions are detected, the service initiates **mobile notification delivery** by sending a payload to **Firebase Cloud Messaging (FCM)**, which in turn pushes the alert directly to the user’s mobile application. To enhance usability, these notifications are equipped with **deep linking**, enabling users to tap on the alert and be taken directly to the relevant forecast details screen within the app. This seamless transition ensures that users can quickly access and act upon critical insights, improving responsiveness and decision-making in real time.

4.4.5 Mobile UI Forecast Details Integration

This component explains how the forecasts generated by the LSTM model are transformed into meaningful insights and presented within the mobile application to make them actionable for farmers. The **Prediction Service (FastAPI)** delivers a structured JSON response designed specifically for mobile use, which includes a **multi-day forecast array** with predictions for the next 5–7 days, each containing the date, predicted price, confidence level, and a trend indicator such as rising, falling, or stable. It also provides **actionable insights** in plain language, derived from model outputs and feature analysis (e.g., “Price is rising likely due to forecasted rain next week”), alongside an **explicit recommendation** such as *DELAY_HARVEST*, *SELL_SOON*, or *NO_CHANGE* to guide farmers in their decision-making. Additionally, the response contains **visualization data**, which combines historical data from the past 30 days with forecasted points, enabling clear graphical representation.

The **mobile application**, built with React Native, includes several UI components to display this data effectively. A **time-series chart**, implemented with libraries such as Victory Native or React Native Charts Wrapper, visually compares historical and forecasted data to highlight price trends. An **insights list** translates the model’s complex outputs into simple bullet points in the local language, ensuring accessibility. A **recommendation banner** is prominently displayed at the top of the screen, presenting the suggested action (e.g., “Delay Harvest by 2 Days”) with intuitive color codes green for positive, red for negative, and blue for neutral. To support rural use cases with unreliable connectivity, an **offline cache** stores the full forecast JSON

object on the device, allowing farmers to view the most recent forecast even without internet access, clearly marked as non-live data.

Finally, the system ensures **seamless alert-to-details navigation**. Push notifications sent via Firebase Cloud Messaging include a **deep link** (e.g., `freshroute://forecast/banana`), which opens the mobile app directly on the forecast details screen for the relevant crop, already pre-populated with the latest data. This streamlined process creates a direct pathway from receiving an alert to accessing comprehensive analysis, enabling farmers to make immediate, informed decisions. In this way, the sophisticated outputs of the LSTM model are not presented as raw predictions, but as clear, visual, and actionable insights tailored to the needs and constraints of rural farmers.

5 SOFTWARE SOLUTION

5.1 Requirement Gathering and Analysis

The requirement-gathering process was done in the form of scrutiny of available research coupled with direct interfacing with stakeholders of Sri Lanka's fruit supply chain to make sure that the solution is evidence-driven as well as problem-driven. Analysis revealed a few prominent pain points that have direct impact on the system design.

Market instability and information asymmetry were also a major issue, as farmers lack timely demand forecasts, thereby often resulting in oversupply, price crashes, and post-harvest losses of 30–40% [1][4]. Middlemen dependence and distorted pricing are also issues, as smallholders are highly reliant on intermediaries who possess market access and exploit farmers using distorted pricing instruments, reducing their earnings [9]. In addition to this, post-harvest and handling losses were also achieved because of inadequate storage facilities, inadequate transportation, and inadequate handling practices, all of which are exacerbated by the absence of cold chain infrastructure. In addition, quality conflicts and mistrust were also achieved because of the absence of standardized quality verification processes, which often result in conflicts between the buyers and farmers and limit the establishment of direct trading relations. These pain areas that were realized influenced not just the functional features, such as AI-driven demand forecasting and quality dispute settlement modules, but also the non-functional features, such as offline-enabled mobile interfaces targeted at rural settings and a growth-enabling cloud-based design. Overall, the solution is specifically designed to digitally bridge such gaps by enabling transparency, predictive insights, and straight-through market linkages, thereby empowering farmers while bridging systemic inefficiencies within the fruit supply chain.

5.2 Project Requirements

5.2.1 Functional Requirements

- ❖ **Demand Forecasting:** The system must generate and update demand forecasts for selected fruits such as bananas, mangoes, and papayas on a daily or weekly basis. Forecasts should use historical price, demand, and weather data, combined with machine learning models like LSTM and Prophet, to predict market demand patterns. This enables farmers to plan harvest and supply more effectively, reducing waste and maximizing profit.
- ❖ **Data Integration:** The system should automatically pull in external data sources, including weather forecasts (rainfall, temperature, humidity) and historical/real-time market prices. This ensures that predictions are up-to-date and account for both climatic conditions and market volatility. Data synchronization must occur at regular intervals (e.g., daily or hourly) without manual intervention, ensuring reliability.
- ❖ **Forecast Dashboard:** A user-friendly mobile interface should display demand curves (trends over time), current weather conditions, and tailored harvest recommendations. The dashboard should use visualizations like graphs and charts to make insights clear and actionable, even for farmers with limited technical literacy.
- ❖ **Alerts & Notifications:** The system should trigger alerts when unusual demand fluctuations are detected. For example, if predicted demand for mangoes suddenly drops below a seasonal average, farmers are notified via SMS and push notifications. Users can also receive early warnings about oversupply, ensuring timely action such as adjusting harvest schedules or diversifying markets.
- ❖ **Offline Operation:** Since many rural areas may have poor connectivity, the mobile app must be able to work offline. Cached forecasts, weather updates, and past alerts should be accessible without internet access. Once connectivity is restored, the system should automatically sync data, ensuring continuity and reliability.
- ❖ **User Management:** The system must allow farmers and buyers to register securely, log in, and manage their profiles. Users should be able to select specific crops of interest, set preferred alert thresholds (e.g., demand increase/decrease percentage), and choose notification methods (SMS, push alerts). This personalization ensures relevance and engagement.
- ❖ **Analytics Interface:** An administrator-facing dashboard should be available to track forecast performance, app usage, and alert effectiveness. This includes monitoring metrics like prediction accuracy, number of active users, frequency of alerts sent, and farmer engagement rates. Such insights help improve the system's accuracy and usability over time.

5.2.2 Non- Functional Requirements

- ❖ **Usability:** The dashboard interface must be intuitive for low-literacy users (using icons, local languages, clear visuals). UI/UX should follow best practices for rural user engagement.
- ❖ **Performance:** Forecast calculations should be completed within a few seconds after data input. The system should handle at least hundreds of users and large datasets efficiently.
- ❖ **Scalability:** The backend architecture must allow for future expansion (e.g., adding more crop types or increased data volume) without major redesign.
- ❖ **Reliability:** Ensure high availability; implement error handling for data fetch failures (e.g., cache last successful data if API fails).
- ❖ **Security:** Protect user data and API keys. All communication should use HTTPS; implement user authentication to secure the dashboard.
- ❖ **Maintainability:** Use modular code and documentation to simplify future updates (e.g., model retraining, UI enhancements).

5.2.3 Expected Test Cases

In the development and implementation of the “FreshRoute” Demand Forecasting module, rigorous testing will be conducted to ensure the accuracy, reliability, and usability of the system. Below is the key test cases expected for this research project.

I. Data Preprocessing and Feature Engineering Test.

Table 2: Test Case 1

Objective	To verify that the system correctly ingests, cleans, and transforms raw data into a suitable format for model training.
Test Case	Provide a dataset with missing values, outliers, and unnormalized numerical features. Verify the preprocessing pipeline handles these issues.
Expected Outcome	Missing values are filled via interpolation, outliers are capped or removed, and all numerical features are normalized to a [0, 1] range. Categorical features like seasonality are correctly encoded.

II. LSTM Model Training and Accuracy Test.

Table 3: Test Case 2

Objective	To validate the predictive accuracy of the LSTM model on historical fruit demand data.
Test Case	Train the model on a preprocessed historical dataset. Use a held-out test set to generate predictions and compare them against actual recorded values.
Expected Outcome	The model achieves a Mean Absolute Percentage Error (MAPE) of less than 15% on the test set, demonstrating sufficient accuracy for actionable forecasting.

III. Real-Time Forecast Generation Test.

Table 4: Test Case 3

Objective	To assess the system's ability to generate a new forecast using the latest market and weather data.
Test Case	Simulate a request to the forecasting API with the most recent 60 days of data. Verifying the output is a valid numerical prediction.
Expected Outcome	The system successfully generates a demand/price forecast for the next day or week and returns it in the specified JSON format via the API.

IV. Alert Triggering and Notification Test.

Table 5: Test Case 4

Objective	To ensure the system triggers alerts when a forecasted value exceeds a user-defined threshold.
Test Case	Set a user's alert threshold for bananas to 10%. Provide a forecast predicting a 15% price increase.
Expected Outcome	The Alert Service successfully triggers and dispatches both an SMS (via Twilio) and a

	push notification (via Firebase) to the user's device.
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V. Mobile Application Functionality Test.

Table 6: Test Case 5

Objective	To validate the usability and core functionality of the React Native mobile application for farmers.
Test Case	Perform key user journeys: logging in, viewing the forecast dashboard, checking detailed forecasts, and managing alert settings. Test with and without an internet connection.
Expected Outcome	The app is intuitive, all features are accessible, and it correctly displays cached forecast data when offline.

VI. System Integration and API Test.

Table 7: Test Case 6

Objective	To evaluate the seamless interaction between the mobile app, backend API, and database.
Test Case	From the mobile app, request a forecast. Trace the request through the API Gateway, to the Forecast Service, and back to the app.
Expected Outcome	The request is handled without errors, and the forecast data is consistently formatted and displayed across all system components.

VII. Performance and Load Testing.

Table 8: Test Case 7

Objective	To evaluate the system's response time and stability under heavy load from multiple users.
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Test Case	Simulate 100+ concurrent users sending forecast requests to the API. Monitor response times and server resource usage.
Expected Outcome	The system maintains stable performance, with all API responses returned in under 2 seconds, demonstrating scalability.

5.3 Feasibility Study

An in-depth feasibility study shows that the proposed demand forecasting system is practical and usable in technical, economic, operational, and scheduling aspects. On the technical side, using common technologies like TensorFlow/Keras for the LSTM model, FastAPI for backend services, React Native for mobile applications, and cloud hosting platforms such as AWS or GCP ensures robust implementation. Historical government datasets from HARTI and the Department of Agriculture in Sri Lanka provide a solid base. However, these datasets need thorough cleaning and preprocessing to address quality issues, which will be handled through effective data cleaning and augmentation.

From the economic standpoint, the system uses open-source technologies to lower software licensing costs. A modest budget of LKR 48,500 is set aside for essential expenses like cloud computing resources, SMS gateway credits from Twilio, and a Google Play Developer account. This expense is reasonable for an end-of-year project and offers a good return on investment through potential agricultural benefits, such as reduced post-harvest losses and increased revenues from better market pricing.

The operational benefits of the system come from its focus on users, featuring a mobile-first design, offline capability, local language support, and multi-channel alerts (smartphone push notifications for smartphone users and SMS for those using feature phones). This approach promotes transparency and encourages adoption among rural farmers while allowing easy integration into current workflows with little change required.

Finally, the project has solid scheduling feasibility, fitting into a 12-month timeline that aligns with the academic calendar. It breaks down into well-defined phases in the Work Breakdown Structure (WBS), including research, design, development, testing, and deployment. Using Agile methods also allows for ongoing improvement and

flexibility to manage risks, such as delays in data collection or model updates. Together, these features ensure the project is viable and capable of delivering effective, scalable solutions for farmers.

5.4 Tools and Technologies

5.4.1 Tools

- ❖ Version Control: Git, GitHub
- ❖ Project Management: Jira
- ❖ Design: Figma (for UI/UX wireframes and prototypes)
- ❖ Model Development Environment: Google Colab
- ❖ IDE: VS Code, PyCharm
- ❖ Communication: Microsoft Teams

5.4.2 Technologies

- ❖ Backend: Python (Flask/Django/FastAPI framework)
- ❖ AI/ML: TensorFlow, Keras, Scikit-learn, Pandas, NumPy
- ❖ Mobile App: React Native (for cross-platform development with offline-first capabilities)
- ❖ Database: PostgreSQL
- ❖ API Integration: RESTful APIs, Requests library
- ❖ Notifications: Twilio API (SMS), Firebase Cloud Messaging (Push Notifications)
- ❖ Deployment: Docker, AWS EC2/Google Cloud Platform/Azure VM (for backend), Google Play Store (for app)

5.5 Implementation

The project will be executed in five key phases. In Phase 1, the focus will be on setting up databases, designing scripts for data collection and preprocessing, and developing LSTM and Prophet models for demand forecasting. Phase 2 involves backend development, including building RESTful APIs such as /api/forecast and /api/alert-preferences, along with implementing the alert generation service. Phase 3 will cover mobile app development, where core app screens like Login, Dashboard, Forecast Details, and Settings will be created, with offline data caching and synchronization

enabled. In Phase 4, all components will be integrated, followed by end-to-end testing that includes model accuracy verification and user acceptance testing (UAT) with a group of farmers. Finally, in Phase 5, the system will be deployed on a cloud server, the mobile app will be published on the Play Store, and monitoring mechanisms will be established to track performance and errors effectively.

5.6 Testing

The testing process will include Unit Testing, where individual components such as data preprocessing functions and API endpoints will be validated using frameworks like Pytest (for Python) and Jest (for React Native). Integration Testing will then ensure smooth interactions between components, for example verifying whether the model correctly saves forecasts to the database and whether the alert service properly triggers the SMS API. Model Validation will focus on assessing the AI models against unseen test data to evaluate predictive accuracy and generalization performance. Finally, User Acceptance Testing (UAT) will involve usability testing with target farmers to collect feedback on the app's interface and functionality, ensuring the system meets user needs effectively.

5.7 Deployment

The backend API and AI services will be containerized using Docker to ensure consistency across environments and deployed on a cloud virtual machine for reliable performance. The mobile application will be published on the Google Play Store, making it accessible for public download. Additionally, environment variables will be utilized to securely manage configurations and sensitive information such as API keys and database passwords, enhancing both flexibility and security in deployment.

6 PERSONAL AND FACILITIES

The project will be executed by a dedicated team of four final-year undergraduate Software Engineering students at the Sri Lanka Institute of Information Technology (SLIIT). The team possesses combined experience in project core domains like artificial intelligence (TensorFlow, time-series forecasting), full-stack web development (Python, FastAPI, React Native), and database management. Academic oversight throughout the project will be provided by the respective research supervisors to guarantee methodological excellence and academic quality.

All development will be performed on the team members' individual laptops, which meet the technical requirements of software development and prototyping machine learning. Google Colab Pro will be utilized for computation-intensive training on the LSTM model using its free GPU. The backend server and database will be running on a cloud service such as AWS, Google Cloud, or the equivalent, leveraging their free tier costs to minimize costs. The other crucial resources, including development environments (VS Code, Python, Git), APIs (Firebase for push notifications, Twilio for SMS), and project management tools (Jira), are all open sources. This strategic plan makes the project financially feasible and can be achieved within the set budget without compromising on technical quality or performance.

7 WORK BREAKDOWN STRUCTURE AND TIMELINE

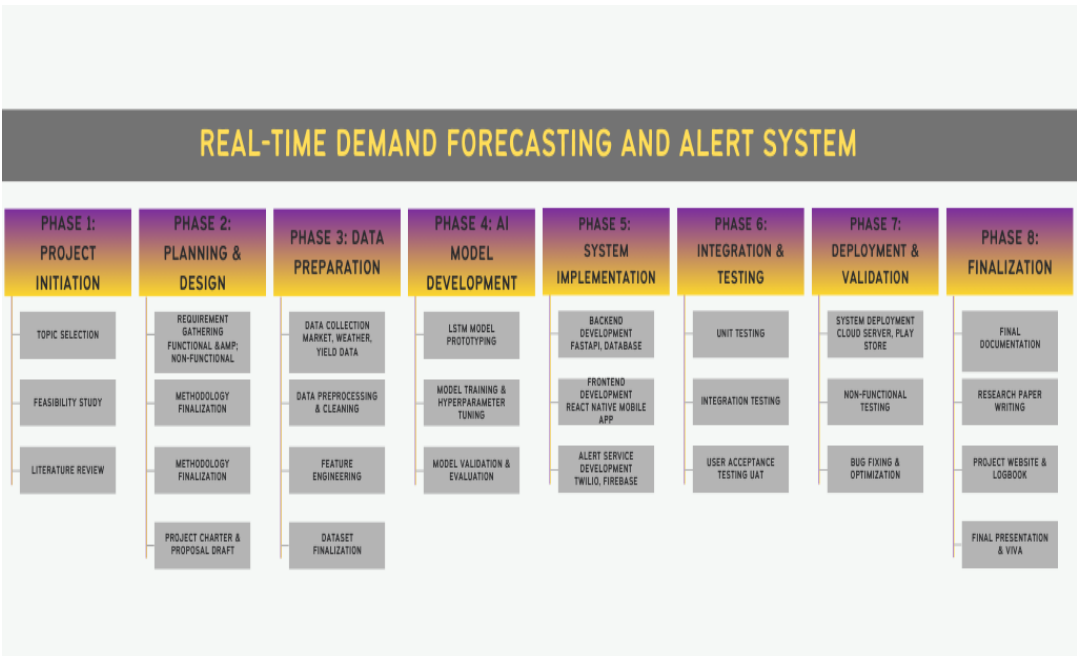


Figure 8: Work Breakdown

This diagram outlines the **phased workflow** of a Real-Time Demand Forecasting and Alert System. It begins with **Project Initiation** (topic selection, feasibility, literature review) and moves to **Planning & Design** (requirements and methodology). **Data Preparation** follows with collection, preprocessing, and feature engineering.

The **AI Model Development** phase focuses on prototyping and validation. **System Implementation** includes backend, mobile app, and alert services. **Integration & Testing** ensures functionality via unit, integration, and user acceptance testing. **Deployment & Validation** covers system deployment, optimization, and non-functional testing. Finally, **Finalization** delivers documentation, research publications, and final presentations, completing the project lifecycle systematically.

8 GANTT CHART

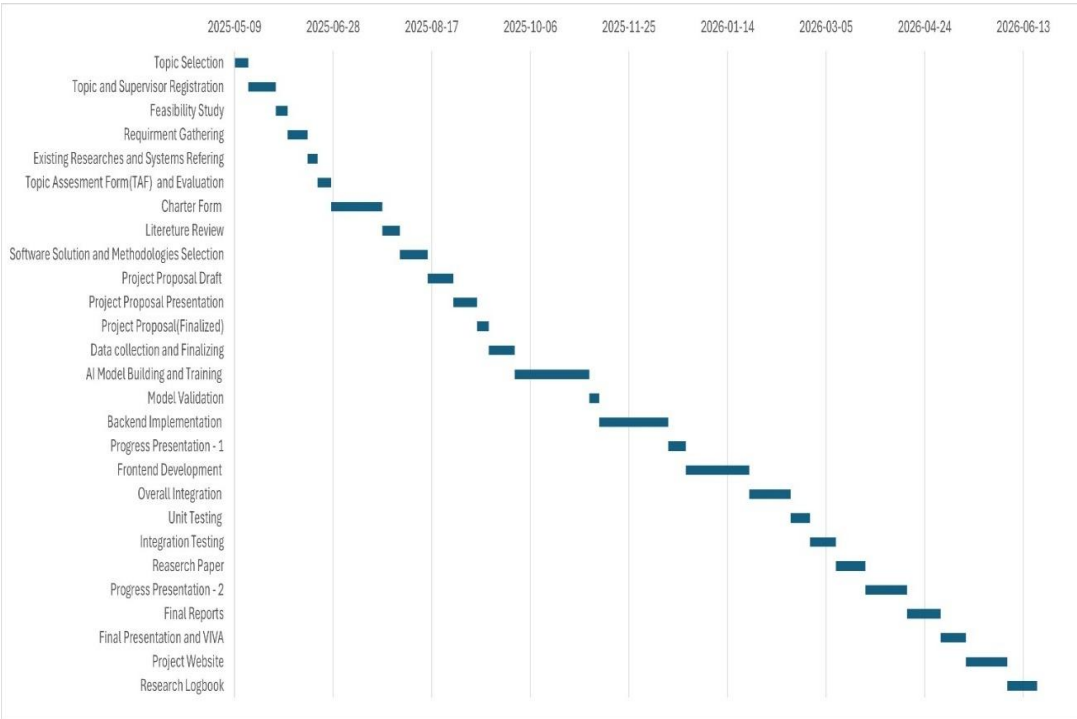


Figure 9: Gantt Chart

9 COMMERCIALIZATION

The demand forecasting module is commercially highly feasible and is envisioned for scalable expansion in Sri Lanka's agricultural sector. The commercialization strategy will pursue a freemium subscription-based model to facilitate rapid adoption among fruit farmers. A free version of the application, designated under the core app, will provide limited features such as short-term forecasts, simple crop price trends, and push notifications [12]. This version will be made available free of charge, lowering the barrier to entry and encouraging widespread use by small-holder farmers.

The principal source of income will be the premium service, FreshRoute Pro, which will provide expert-level features such as long-term market projections, multi-crop forecasts, seasonal trends, and priority SMS alerts. Such value-added services will have appeal for medium- and large-scale farmers requiring more detailed information to plan harvest cycles, negotiate prices with buyers, and reduce post-harvest losses. For maximum coverage, strategic partnerships with telecommunication providers like Mobitel and Dialog will be pursued, enabling bundled SMS and data packages that reduce farmers' cost of operations. Partnerships with the Ministry of Agriculture and allied government departments can also provide credibility, subsidies, and alignment with national farmer welfare initiatives. Partnerships with financial institutions will also enable value-added services like linking forecasts to microloans and crop insurance, which can increase farmer confidence and adoption.

The initial target market will be Sri Lanka's key fruit-growing regions, where the demand-supply imbalance is most evident. Following adoption, the service will be offered to vegetable and spice growers, broadening the user base [13]. Marketing will emphasize on-ground demonstrations, training through local agents, and regional language campaigns localized in nature, ensuring inclusiveness.

The long-term vision is to develop the platform into an agricultural intelligence ecosystem. Through anonymized and aggregated data, the system will offer insights valuable to policymakers, exporters, and agribusinesses. This not only generates diversified revenue streams but also assists in making the agricultural value chain in Sri Lanka more efficient, transparent, and sustainable.

10 BUDGET AND BUDGET JUSTIFICATION

Table 9: Budget Table

Description	Amount (LKR)
Web hosting	3000.00
Model Training Resources	12000.00
SMS Gateway Credits (Twilio/Local)	5000.00
Data Collection & APIs	5000.00
Mobile Development tools cost	10000.00
Internet cost	3000.00
Cloud services	5000.00
Miscellaneous	10000.00
Total Estimated Budget	53000.00

Justification:

This budget covers the essential operational costs required to develop, test, and deploy a fully functional prototype of the FreshRoute system. The largest expenses are for Model Training Resources (LKR 12,000) to ensure the AI forecast is accurate and reliable, and SMS Gateway Credits (LKR 5,000) to provide critical alerts to farmers with basic phones. Costs for Web Hosting, Cloud Services, and APIs (LKR 13,000 combined) are necessary for a stable and scalable backend infrastructure. The Miscellaneous fund (LKR 10,000) is allocated for unforeseen expenses, transportation for field testing with farmers, and other incidentals, ensuring the project can be completed smoothly without compromising on quality.

11 LIMITATIONS, ASSUMPTIONS AND CHALLENGES

Data limitations are significant: historical market and weather data may be incomplete or noisy, which can reduce forecast accuracy. Advanced models like LSTM require careful tuning; as noted in prior work, even LSTM models can be difficult to fine-tune across diverse datasets [6],[7],[8]. Consequently, we must iteratively refine the models and possibly simplify if convergence issues arise.

Connectivity and adoption pose another challenge. Rural farmers often have limited or intermittent internet access. Although the app is designed for offline use, this means forecasts can only be updated when connectivity is available. Additionally, farmers may not frequently check new apps or SMS alerts [11]. We will mitigate this through user-centered design: minimal text, clear icons, and minimally required user actions.

Technical constraints include SMS reliability and cost: message delivery can be delayed or fail (e.g., if a phone is switched off), and SMS credits limit alert frequency. Finally, model limitations (e.g., unexpected market shocks, pest outbreaks) cannot be fully predicted by historical patterns. We will manage expectations by supplementing forecasts with explanations and encouraging farmers to use the alerts as guidance rather than absolute predictions.

By acknowledging these challenges, we can plan risk mitigation strategies (e.g., data validation, user training) to improve system robustness.

12 CONCLUSIONS

The FreshRoute project conclusively demonstrates the viability of an AI-driven solution to tackle critical inefficiencies in Sri Lanka's fruit supply chain. By developing a sophisticated LSTM-based forecasting model, this research successfully integrates historical market data, weather patterns, and seasonal trends to generate accurate, real-time demand predictions. The system directly addresses the core problem of information asymmetry by delivering these actionable insights directly to farmers via an offline-capable mobile application and multi-channel SMS alerts, ensuring accessibility even in low-connectivity rural areas. This empowers farmers to make informed decisions on optimal harvest timing, thereby minimizing post-harvest waste and maximizing profitability by aligning supply with market demand. The project fulfills its objectives by not only providing a technically sound and validated forecasting module but also by designing a practical, user-centric tool that bridges the gap between advanced technology and end-user capability. While challenges such as data quality and model adaptability remain, the proposed system lays a strong foundation for enhancing supply chain resilience, reducing reliance on middlemen, and improving overall food security. Future work will focus on model refinement, large-scale pilot testing, and exploring commercial sustainability, ultimately aiming to transform agricultural decision-making for Sri Lankan fruit farmers.

REFERENCES

- [1] L. N. C. De Silva, J. S. Goonetillake, G. N. Wikramanayake, and A. Ginige, “A holistic mobile-based information system to enhance farming activities in Sri Lanka,” in *Proc. IASTED Int. Conf. Engineering and Applied Science (EAS)*, Dec. 2012, pp. 91–99. [Online]. Available: <https://www.researchgate.net/publication/261367287>
- [2] M. Aphinaya, R. Rathnayake, S. Sivakumar, and A. M. C. Amarakoon, “Price forecasting of jack fruit using SARIMA model,” Univ. of Jaffna, Sri Lanka, 2016. [Online]. Available: <http://repo.lib.jfn.ac.lk/ujrr/handle/123456789/2028>
- [3] T.-W. Yoo and I.-S. Oh, “Time series forecasting of agricultural products’ sales volumes based on seasonal long short-term memory,” *Appl. Sci.*, vol. 10, no. 22, art. 8169, 2020. [Online]. Available: <https://www.researchgate.net/publication/346743222>
- [4] H. M. B. P. Ranaweera, R. M. G. H. N. Rathnayake, and A. S. G. J. K. Ananda, “Crop price prediction using machine learning approaches: Reference to the Sri Lankan vegetable market,” *J. Manage. Matters*, vol. 10, no. 1, pp. 19–34, 2023. [Online]. Available: <https://jmm.sljol.info/articles/10.4038/jmm.v10i1.7670>
- [5] N. S. SriLakshmi, K. Pavan, K. Divya, P. Swetha, and G. Padmaja, “Integrated agricultural monitoring system with SMS alerts and web application,” *Int. J. Res. Trends Innov.*, vol. 9, no. 3, p. 59, 2024. [Online]. Available: <https://www.ijrti.org/viewPaper.php?paperId=IJRTI2403007>
- [6] Y. H. Gu, Z. L. Yang, X. H. Zhang, and Y. Chen, “Forecasting agricultural commodity prices using dual input attention LSTM,” *Agriculture*, vol. 12, no. 2, art. 256, 2022. [Online]. Available: <https://www.mdpi.com/2077-0472/12/2/256>
- [7] R. L. Manogna, V. Dharmaji, and S. Sarang, “Enhancing agricultural commodity price forecasting with deep learning,” *Sci. Rep.*, vol. 15, art. 20903, 2025. [Online]. Available: <https://www.nature.com/articles/s41598-025-20903>
- [8] T. Xiong, C. Li, and Y. Bao, “Seasonal forecasting of agricultural commodity price using a hybrid STL and ELM method: Evidence from the vegetable market in China,” *Neurocomputing*, vol. 275, pp. 2831–2844, Jan. 2018.
- [9] P. Wanigasinghe and D. D. M. Ranasinghe, “Price prediction neural network for Econ Centre,” in *Proc. 9th Annu. Sessions Sri Lanka Assoc. Artif. Intell. (SLAAI)*, The Open University of Sri Lanka, Dec. 2012, pp. 1–8.
- [10] D. I. Yoo, “Developing vegetable price forecasting model with climate factors,” *Korean J. Agric. Econ.*, vol. 57, no. 1, pp. 1–24, 2016.

- [11] M. Fafchamps and B. Minten, "Impact of SMS-based agricultural information on Indian farmers," *World Bank Econ. Rev.*, vol. 26, no. 3, pp. 383–414, 2012.
- [12] M. A. N. S. Perera, P. K. Liyanage, and K. S. S. Perera, "Digital transformation in agriculture: Business models and commercialization strategies for ICT-based solutions in Sri Lanka," *ICTer Journal*, vol. 15, no. 2, pp. 42–56, Dec. 2022. [Online]. Available: <https://icter.org/journal/vol15/issue2/>
- [13] A. Klerkx, E. Jakku, and P. Labarthe, "A review of social science on digital agriculture, smart farming and agriculture 4.0: New contributions and a future research agenda," *NJAS - Wageningen Journal of Life Sciences*, vol. 90–91, 2019, art. 100315. [Online]. Available: <https://doi.org/10.1016/j.njas.2019.100315>

APPENDIX

 දඹුල්ල විශේෂිත ආර්ථික මධ්‍යස්ථාන කළමනාකරණ භාරය ද.අ: -0662285181 Email : dambulladec@gmail.com 					
දෛනික මිල තොරතුරු, තොග මිල : (R&K)					
දිනය : 2025.08.27		වේලාව : 9.30 am			
එළවළු වර්ග		මිල පරාසය		අලු වර්ග	
ගෝවා - Cabbage	160.00	200.00	ගොවම් අල - Lotus Roots	450.00	500.00
මල්ගෝවා - Cauliflower	250.00	320.00	ගොහිල අල - Kohila	100.00	110.00
රතුගෝවා - Red Cabbage			කිරි අල - Kiriala	150.00	170.00
බෝංචි - Beans	250.00	400.00	මස්සොක්කා - Manioc	70.00	80.00
ලීක්ස් - Leeks	90.00	120.00	බතල - Sweet Potato	100.00	120.00
කැරට් - කුට්ටිමිල - Carrot - N	200.00	280.00	ඉන්තල - Wild - Baby Potato		
කැරට් - යාපනය - Carrot - Jaffna					
ගොළුබීට් - කුට්ටිමිල - Beetroot N	80.00	100.00	පලාචර්ග		
කටබීට් - මැල්බීට්සුට් - Beetroot Cut - M	60.00	70.00	ලුණුකොළ - Onion Leaves	140.00	150.00
කටබීට් - දැල්ල - Beetroot Cut - D	60.00	65.00	මුතුකුඩුවන්නා - Mukunuwenna	20.00	25.00
කටබීට් - යාපනය - Beetroot Cut - J	30.00	50.00	සලාද - Lettuce	200.00	250.00
කටබීට් - සුන්තලම - Beetroot Cut - P	70.00	80.00	ගොටුකොළ - Gotukola	20.00	25.00
තක්කාලි - Tomato	120.00	170.00	කරපි-වා - Curry Leaves	60.00	70.00
රාබු - Raddish	30.00	40.00	ඉතුගෝවා - Athugowa	100.00	150.00
නෝකෝල් - Nnolkhol	60.00	70.00	හිටිති - Spinach	60.00	65.00
කැකිට් - Cooking Melon	30.00	40.00	ක-කුන් - Water Spinach	20.00	25.00
සිසිල්ලා - Cucumber	70.00	80.00	බාහාය		
අරිකාසල් - කුට්ටිමිල - Potatoes N	230.00	250.00	බඩ ඉරිඟු ඇට - Corns	110.00	120.00
අරිකාසල් - යාපනය - Potatoes J			සෝයා බෝංචි - Soya Bean	290.00	300.00
අරිකාසල් - වැලිමිල - Potatoes W			සියඹිලා - Tamarind	490.00	500.00
අරිකාසල් - ආනයන - Potatoes Import	170.00	180.00	සල - Sesame	350.00	500.00
චැට්කුළු - Ridge Gourd	170.00	200.00	හී-ඇට - Green Gram	590.00	600.00
බණ්ඩක්කා - Lady's Fingers	70.00	100.00	කඩපි - Cowpea	590.00	600.00
පනෝල - Snake Gourd	100.00	130.00	කරක්කන් - Finger Millet	390.00	400.00
කරවිල - Bitter Gourd	200.00	220.00	කඩල - Gram	350.00	360.00
තුම්කරවිල - Thumbakarawila	450.00	550.00	ආඤ්ඤා - Peanuts	490.00	500.00
වට්ටක්කා - මැල්මිල - Pumpkin	30.00	35.00	පළතුරු වර්ග		
වට්ටක්කා - ලොකු - Pumpkin	50.00	60.00	බීති කෙසෙල් - Banana - Sini	60.00	70.00
බට්ටානා - Butternut Squash	50.00	55.00	හුඹුල් කෙසෙල් - Banana - Ambul	85.00	90.00
වම්බු - Brinjal	300.00	350.00	කොලිකුට්ටු - Kolikuttu	210.00	220.00
තිත්බටු - Thithbatu	300.00	350.00	ඇම්බුන් - Ambun Banana	160.00	170.00
චුචු - Thai Eggplant	250.00	300.00	ජකඹිලා කෙසෙල්	290.00	300.00
බෝංචි - Eggplant	220.00	230.00	ඉජර - Guava	270.00	280.00
දිවුල් - Winged Bean	200.00	230.00	පැපායා - Papaya	85.00	90.00
මැකරල් - Yard - Long Beans	100.00	200.00	කොමඩු - Watermelon	85.00	90.00
ඩ්‍රම්ස්ටික්ස් - Drumsticks	150.00	200.00	පින්නාඩ් - Pineapple	270.00	280.00
අඬුකෙසෙල් - Ash Plantain	60.00	70.00	චාළු අඬු	130.00	140.00
කෙසෙල්බුට් - Plantain Flower	50.00	60.00	කරන්ත කොළොච්චා	540.00	550.00
ඇම්බර්ලා - Ambarella	50.00	60.00	ඇල්පොන්සු - TJC	840.00	850.00
චුචු - Chow - Chow	50.00	60.00	චුචුකන් අඬු - Mango	240.00	250.00
ලයිම් - Lime	450.00	550.00	පැපෝන් - Passion	410.00	420.00
මාළුමිරිස් - Capsicum	200.00	300.00	ඇවිඳේර - Avocado	190.00	200.00
අඹුමිරිස් - Green Chili	120.00	200.00	කටු අනෝසා - Soursop	210.00	220.00
වියළි මිරිස් - Dry Chilies	580.00	590.00	චුචුකන් කොල්ලි - Gooseberry	1,200.00	1,300.00
නයිමිරිස් - Nai Miris	700.00	1,000.00	චුචුල් - Woodapple	190.00	200.00
ඉඟුරු - Genger	850.00	900.00	චුචුල් - Indian bael	340.00	350.00
කහ - Turmeric	240.00	250.00	කැකිලි - King Coconut	120.00	130.00
පොළොස් - Jackfruit	70.00	80.00	පොල් - Coconut	130.00	140.00
රතුරුණු - ආනයන - Red Onion Import					
රතුරුණු - ලංකා - Red Onion - Lanka	130.00	200.00			
ලොකුරුණු - ආනයන - B Onion Import	120.00	130.00			
ලොකුරුණු - ලංකා - B Onion Lanka					
සලාද - Garlic	340.00	350.00			